Noether Identities and Discrete φ-Embedding in Compact S^3 Quantum Cosmology

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We present a topology-explicit and Noether-consistent variational framework for quantum cosmology on a spacetime $M = \mathbb{R} \times \Sigma$ with compact spatial hypersurfaces Σ, and we make explicit the topological sector in which the universal cover is spherical, $\widetilde{\Sigma} \simeq S^3$. General covariance implies a Noether-II identity structure (Bianchi-type identities) rather than genuine global conserved charges. We then introduce a discrete φ-embedding level $n \in \mathbb{Z}$ (or a time-dependent embedding $N(t)$) within the action and state precise conditions under which an on-shell conserved current $\nabla_\mu J^\mu_{(n)} = 0$ exists. The discrete choice $N(t) = \lfloor \log_\varphi(ct/\ell_P) \rfloor$ is shown to require explicit junction conditions at transition times to prevent distributional sources in the Euler–Lagrange equations. Alternatively, a smooth embedding variable $\nu(t) = \log_\varphi(ct/\ell_P)$ yields a fully differentiable action, with discretization emerging at the level of spectra. We derive the boundary/topology corollary: for compact Σ with $\partial\Sigma = \varnothing$, diffeomorphism-Noether charges reduce to boundary integrals and vanish globally (in particular for $\Sigma \cong S^3$). Finally, we provide minimal, falsifiable observational discriminants in CMB anisotropies and stochastic gravitational-wave backgrounds sensitive to compact topology and discrete mode structure, and we outline a likelihood-based verification pipeline.

I. MOTIVATION AND SCOPE

In generally covariant theories such as General Relativity (GR), the relevant symmetry is local: invariance under diffeomorphisms $\mathrm{Diff}(M)$. Noether's second theorem [1] therefore applies: the symmetry implies differential identities among the Euler–Lagrange (EL) expressions rather than independent global conservation laws of Noether-I type. This distinction becomes operationally critical in cosmology, where global notions such as "total energy of the Universe" depend sensitively on the existence of boundaries and the global topology of spatial hypersurfaces Σ.

The aim of this manuscript is twofold: (i) to state the Noether-II content of GR in a topology-explicit way (boundary charges and their vanishing on compact boundaryless Σ with $\partial\Sigma = \varnothing$, including the spherical sector $\widetilde{\Sigma} \simeq S^3$), and (ii) to formulate a mathematically consistent implementation of a discrete φ-embedding label n (or $N(t)$) inside a variational principle, including the required junction conditions for piecewise-smooth embeddings. We then connect the framework to observational discriminants of compact topology: discrete Laplacian spectra, low-ℓ CMB structure, and infrared properties of the stochastic gravitational-wave background.

A. Invariant fixing of the golden ratio scale factor

The discrete embedding label n is not introduced as a tunable convention, but as an invariant bookkeeping of *scale composition*: coarse-graining steps must add without ambiguity, and the resulting representation of integers must be unique under refinement. The mathematically sharp form of this requirement is the *Zeckendorf property*: every positive integer admits a unique decomposition into a sum of *non-adjacent* elements of a strictly increasing "scale sequence." In the classical case this sequence is the Fibonacci ladder, and uniqueness is guaranteed by the Zeckendorf–Lekkerkerker theorem [33, 34].

Physically, this "unique decomposition" is the minimal invariance needed for a well-defined refinement semigroup: successive coarse-grainings must compose associatively and yield a path-independent index of scale. In quantum cosmology this provides an invariant coarse-graining coordinate (a discrete RG-like clock) rather than an arbitrary choice of base.

To avoid confusion with the large "golden-ratio cosmology literature (often based on post-hoc pattern matching), we stress that φ does not enter here as a fit parameter or an empirical coincidence. Instead, φ is a *rigidity consequence* of imposing a minimal invariance requirement on scale composition (unique, path-independent refinement in a discrete semigroup) *inside* the variational principle. The empirical content of the framework is therefore not "φ appears in data, but rather the falsifiable prediction that compact-topology observables must be computed with discrete mode sums and (if present) φ-indexed features, and then tested against likelihoods (Secs. V H and VI).

A key rigidity statement (a weak converse) shows that this uniqueness requirement is not merely consistent with the Fibonacci ladder but essentially characterizes it among monotone sequences [35]. Hence the discrete scale ladder is fixed (up to initial normalization) by invariance/uniqueness alone. The associated *scale multiplier* is then forced as the unique positive fixed point of the induced recursion:

Lemma 1 (Golden ratio as the invariant multiplier of the Zeckendorf ladder)**.** *Let $\{F_n\}$ be the Fibonacci sequence defined by $F_{n+2} = F_{n+1} + F_n$ with $F_0 = F_1 = 1$. Then the ratio $r_n = F_{n+1}/F_n$ converges to the unique positive root of $r^2 = r + 1$, namely*

$$\lim_{n\to\infty} \frac{F_{n+1}}{F_n} = \varphi \equiv \frac{1+\sqrt{5}}{2}. \tag{1}$$

In the present framework, n labels this invariant discrete ladder of scale composition; therefore φ enters as the uniquely selected multiplier of the ladder, and the smooth embedding $\nu(t) = \log_\varphi(ct/\ell_P)$ is the coordinate-free continuum interpolation of that invariant index.

II. GEOMETRIC SETUP AND VARIATIONAL CALCULUS

A. Spacetime and topology

Definition 1 (Spacetime decomposition)**.** Let $M = \mathbb{R} \times \Sigma$, where Σ is a compact orientable 3-manifold *without boundary* ($\partial\Sigma = \varnothing$). The Noether-II/topology statements below depend only on this boundarylessness.

Proposition 1 (Spherical cover as an invariantly characterized sector)**.** *If Σ admits a (complete) Riemannian metric of constant* positive *sectional curvature, then Σ is a spherical space form: there exists a finite group $\Gamma \subset \mathrm{Isom}(S^3)$ acting freely such that*

$$\Sigma \simeq S^3/\Gamma, \tag{2}$$

and hence the universal cover is $\widetilde{\Sigma} \simeq S^3$.[12, 13]

In the observational part of the paper, this spherical-cover sector is singled out by its invariant spectral fingerprints (discrete Laplacian eigenvalues and characteristic degeneracies), which are directly testable in low-ℓ CMB statistics and in infrared features of $\Omega_{\mathrm{GW}}(f)$.

B. Action and EL expressions

Consider a generally covariant action

$$S[g,\psi] = \int_M \mathrm{d}^4x\,\sqrt{-g}\,\mathcal{L}(g,\psi,\nabla\psi,\nabla\nabla\psi,\dots), \tag{3}$$

where ψ denotes matter or effective cosmological degrees of freedom. We define EL expressions as

$$E^{\mu\nu} \equiv \frac{1}{\sqrt{-g}}\frac{\delta S}{\delta g_{\mu\nu}}, \qquad E_\psi \equiv \frac{1}{\sqrt{-g}}\frac{\delta S}{\delta \psi}. \tag{4}$$

III. NOETHER II FOR DIFFEOMORPHISM INVARIANCE

A. Local symmetry and identities

Theorem 1 (Noether II for diffeomorphism invariance)**.** *Let $S[g,\psi]$ be invariant under infinitesimal diffeomorphisms generated by an arbitrary smooth vector field $\xi^\mu(x)$:*

$$\delta_\xi g_{\mu\nu} = \nabla_\mu\xi_\nu + \nabla_\nu\xi_\mu, \qquad \delta_\xi\psi = \mathcal{L}_\xi\psi. \tag{5}$$

Then the EL expressions satisfy differential identities (Noether-II identities) of the schematic form

$$\nabla_\mu E^{\mu\nu} \;+\; \mathcal{K}^\nu(E_\psi, \nabla E_\psi, \dots) \;\equiv\; 0, \tag{6}$$

where $\mathcal{K}^\nu$ is a covariant differential operator determined by the tensor character of ψ.

Proof. Diffeomorphism invariance depends on arbitrary functions $\xi^\mu(x)$, hence it falls under Noether's second theorem [1]. Varying S under (5) yields the schematic identity

$$\begin{aligned} 0 = \delta_\xi S = &\int_M \mathrm{d}^4x\,\sqrt{-g}\Big(E^{\mu\nu}\delta_\xi g_{\mu\nu} + E_\psi\,\delta_\xi\psi\Big) \\ &+ \int_M \mathrm{d}^4x\,\sqrt{-g}\,\nabla_\mu(\cdots). \end{aligned} \tag{7}$$

Integrating by parts transfers derivatives from ξ^μ to $E^{\mu\nu}$ and E_ψ. Since $\xi^\mu(x)$ is arbitrary, the coefficient of ξ^μ must vanish identically, giving (6). □

B. Boundary charges, compact Σ, and the meaning of "conservation"

In GR, diffeomorphism invariance is a local gauge symmetry. Consequently, the standard Noether-I expectation ("rigid symmetry ⇒ a global conserved charge") does not apply in a straightforward way. Noether's second theorem instead yields identity structure among the Euler–Lagrange expressions, and the corresponding "conservation laws" are improper in the classic sense: on-shell they reduce to total divergences and/or boundary terms rather than providing a unique local, gauge-invariant gravitational energy density [1, 2].

A precise formulation is given by the covariant phase space approach [3–5]. Treat the Lagrangian as an n-form L and write

$$\delta L \;=\; E(\phi)\,\delta\phi \;+\; d\Theta(\phi,\delta\phi), \tag{8}$$

where $E(\phi) = 0$ are the field equations and Θ is the symplectic potential. For a vector field ξ generating an infinitesimal diffeomorphism, define the Noether current $(n-1)$-form

$$J_\xi \;\equiv\; \Theta(\phi,\mathcal{L}_\xi\phi) \;-\; \xi\cdot L. \tag{9}$$

Diffeomorphism invariance implies that J_ξ is closed on-shell, $dJ_\xi \approx 0$, and, crucially, on-shell *exact*:

$$J_\xi \;\approx\; dQ_\xi, \tag{10}$$

with Q_ξ an $(n-2)$-form (Noether charge form). This is the clean statement that covariant charges in GR are generically boundary charges.

At the level of canonical generators, one may write schematically

$$H_\xi \;=\; \int_{\partial\Sigma}\big(Q_\xi - \xi\cdot B\big), \tag{11}$$

where B is a boundary counterterm fixed by the admissible boundary conditions [4, 5]. Equation (11) makes the compact-sector corollary immediate.

Corollary 1 (Compact boundaryless Σ)**.** *If Σ is compact and boundaryless, $\partial\Sigma = \varnothing$, then diffeomorphism-Noether charges reduce to boundary integrals and vanish globally:*

$$\partial\Sigma = \varnothing \qquad \Longrightarrow \qquad H_\xi = 0. \tag{12}$$

(For the corresponding ξ and standard boundary conditions.) In particular, for $\Sigma \cong S^3$ (or more generally $\Sigma \simeq S^3/\Gamma$), there is no global Noether-I "energy of the Universe"; the invariant content is the Noether-II identity structure in Theorem 1.

Remark 1 (Junction consistency under discrete refinement). When refinement is implemented piecewise (e.g. via the floor embedding $N(t)$ below), Noether-consistency requires that matching data do not introduce spurious distributional sources. In the covariant language this means that the exactness property (10) is not spoiled by junction contributions beyond admissible boundary counterterms. This is the conceptual origin of the explicit junction conditions derived in Section IV.

IV. DISCRETE φ-EMBEDDING: SMOOTH VS FLOOR AND JUNCTION CONDITIONS

A. Embedding variables

Define the smooth embedding variable

$$\nu(t) = \log_\varphi\left(\frac{c\,t}{\ell_P}\right), \tag{13}$$

and the discrete embedding level

$$N(t) = \lfloor \nu(t) \rfloor, \qquad N \in \mathbb{Z}. \tag{14}$$

B. Prototype φ-dependent matter sector

As a minimal prototype, consider a complex scalar ψ with a φ-dependent potential:

$$\mathcal{L} = \frac{1}{2} g^{\mu\nu}\partial_\mu\psi^*\partial_\nu\psi - V(|\psi|; N),$$
$$V(|\psi|; N) = \frac{1}{2} m^2 |\psi|^2\, \varphi^{-2N}. \tag{15}$$

More generally, one may encode the embedding level as a conformal scaling

$$g_{\mu\nu}(x, N) = \varphi^{-2N}\, \tilde{g}_{\mu\nu}(x), \tag{16}$$

provided the physical (nondegenerate) metric entering observables is defined so that degeneracy is avoided. Concretely, one may treat $\tilde{g}_{\mu\nu}$ as the physical metric and interpret (16) as an *auxiliary* scale bookkeeping that is absorbed into the definition of dimensionless couplings and potentials; then all observables are built from dimensionless ratios and remain well-defined in the smooth limit.

C. Two consistent formulations

a. (I) Smooth embedding.

$$S = \int \mathrm{d}^4x\, \sqrt{-g}\, \mathcal{L}(\phi, \nabla\phi; \nu(t)). \tag{17}$$

b. (II) Discrete floor embedding.

$$S = \int \mathrm{d}^4x\, \sqrt{-g}\, \mathcal{L}(\phi, \nabla\phi; N(t)). \tag{18}$$

D. Junction term from the variational principle

Splitting into intervals $I_k = (t_k, t_{k+1})$ and integrating by parts yields

$$\delta S = \sum_k \int_{I_k} \mathrm{d}^4x\, \sqrt{-g}\, \Big(E_\phi\, \delta\phi\Big) + \sum_k \int_{\Sigma_{t_k}} \mathrm{d}^3x\, \left[\Pi_\phi\, \delta\phi\right]_{t_k^-}^{t_k^+}, \tag{19}$$

with

$$\Pi_\phi \equiv \frac{\partial(\sqrt{-g}\mathcal{L})}{\partial(\partial_t\phi)}. \tag{20}$$

Theorem 2 (Junction consistency for distribution-free EL equations)**.** *In the discrete formulation* (18)*, the EL equations are distribution-free iff*

$$\int_{\Sigma_{t_k}} \mathrm{d}^3x \left[\Pi_\phi\, \delta\phi\right]_{t_k^-}^{t_k^+} = 0 \qquad \textit{for all } k. \tag{21}$$

A sufficient standard choice is continuity of canonical data:

$$\phi(t_k^-) = \phi(t_k^+), \qquad \Pi_\phi(t_k^-) = \Pi_\phi(t_k^+). \tag{22}$$

Proof. (19) is exact. Stationarity requires both the bulk and junction contributions to vanish for all admissible variations, giving (21). □

E. Implication for conserved currents

If on each I_k there is a current $J^\mu_{(N)}$ with $\nabla_\mu J^\mu_{(N)} = 0$ on-shell, then global conservation holds iff

$$\sum_k \int_{\Sigma_{t_k}} \mathrm{d}^3x\, \sqrt{h}\, n_\mu J^\mu_{(N)}\Big|_{t_k^-}^{t_k^+} = 0. \tag{23}$$

F. Explicit generalized action and scaling symmetry (comparison to the two preprints)

For completeness, and to make the present manuscript self-contained, we restate the core structures of the two companion preprints on a "generalized Noether theorem" [6, 7] in the FBS^{3}R setting (quantum-cosmology and multifractal versions) in a unified notation.

a. (A) Fixed embedding level n (quantum-cosmology sketch). The preprint model starts from a φ-indexed family of length scales

$$R_n = \ell_{\rm P}\,\varphi^{\,n}, \tag{24}$$

and an action for a complex scalar "wave function of the Universe" ψ on a background metric $g_{\mu\nu}$,

$$S_n[\psi, g] = \int d^4x\,\sqrt{-g}\;\mathcal{L}(\psi, \partial_\mu\psi; \varphi^{\,n}), \tag{25}$$

with Lagrangian density of the form

$$\mathcal{L} = \frac{1}{2}\,g^{\mu\nu}\,\partial_\mu\psi^*\,\partial_\nu\psi - V(|\psi|, \varphi^{\,n}),$$
$$V(|\psi|, \varphi^{\,n}) = \frac{1}{2}\,m^2|\psi|^2\,\varphi^{-2n}. \tag{26}$$

The generalized scaling transformation is taken as

$$x^\mu \mapsto \varphi\,x^\mu, \tag{27}$$

and the associated Noether current (in the standard variational form) is

$$J^\mu_{(n)} = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi)}\,\delta\psi - \Big(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi)}\,\partial_\nu\psi - \mathcal{L}\,g^\mu{}_\nu\Big)\delta x^\nu. \tag{28}$$

On shell this satisfies the covariant conservation law

$$\nabla_\mu J^\mu_{(n)} \approx 0, \tag{29}$$

where $\approx$ denotes equality modulo the Euler–Lagrange equations.

b. (B) Multifractal action with a time-dependent embedding level $N(t)$. The multifractal preprint specifies a metric scaling

$$ds^2 = g_{\mu\nu}(x, n)\,dx^\mu dx^\nu, \qquad g_{\mu\nu}(x, n) = \varphi^{-2n}\,\tilde{g}_{\mu\nu}(x), \tag{30}$$

and a variational principle

$$S = \int d^4x\,\sqrt{-g}\;\mathcal{L}(\psi, \partial_\mu\psi, n). \tag{31}$$

If $n = n(\psi)$ is treated as an additional dependent variable, the Euler–Lagrange equation takes the modified form

$$\frac{\partial\mathcal{L}}{\partial\psi} - \nabla_\mu\Big(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi)}\Big) + \frac{\partial\mathcal{L}}{\partial n}\,\frac{\partial n}{\partial\psi} = 0. \tag{32}$$

The embedding level is then promoted to a time-dependent quantity

$$N(t) = \Big\lfloor \log_{\frac{1+\sqrt{5}}{2}}\Big(\frac{ct}{\ell_{\rm P}}\Big)\Big\rfloor = \Big\lfloor \log_{\varphi}\Big(\frac{ct}{\ell_{\rm P}}\Big)\Big\rfloor, \tag{33}$$

and the action is written as

$$S_{\rm FBS^3R} = \int d^4x\,\sqrt{-g_{N(t)}}\;\mathcal{L}_{N(t)}\Big(\psi, \partial_\mu\psi, \varphi^{N(t)}\Big), \tag{34}$$

with

$$\mathcal{L}_n = \frac{1}{2}\,g^{\mu\nu}\partial_\mu\psi^*\partial_\nu\psi - V\big(|\psi|, \varphi^n\big),$$
$$V\big(|\psi|, \varphi^n\big) = \frac{1}{2}\,m^2|\psi|^2\,\varphi^{-4n}. \tag{35}$$

The preprint further proposes an illustrative two-regime ansatz for the wave function,

$$\psi(t) = \begin{cases} A\left(\dfrac{ct}{\ell_{\rm P}}\right)^{-3/2}\exp\Big(i\,\varphi^{N(t)}\,\dfrac{t}{t_{\rm P}}\Big), & t < t_{\rm rec}, \\ B\,\exp\big(-\frac{3}{2}H(t - t_{\rm rec})\big)\,\exp(iHt), & t \geq t_{\rm rec}, \end{cases} \tag{36}$$

In the "large-N" continuum approximation used for intuition in the multifractal sketch, one writes

$$\exp\Big(i\,\varphi^{N(t)}\frac{t}{t_{\rm P}}\Big) \;\sim\; \exp\Big(i\,\frac{ct}{\ell_{\rm P}}\,\ln t\Big), \tag{37}$$

$$g_{\mu\nu}(x, N(t)) = \varphi^{-2N(t)}\tilde{g}_{\mu\nu}(x) \;\sim\; \left(\frac{ct}{\ell_{\rm P}}\right)^{-2}\tilde{g}_{\mu\nu}(x), \tag{38}$$

$$V\big(|\psi|, \varphi^{N(t)}\big) = \frac{1}{2}m^2|\psi|^2\varphi^{-4N(t)} \;\sim\; \frac{1}{2}m^2|\psi|^2\left(\frac{ct}{\ell_{\rm P}}\right)^{-4}, \tag{39}$$

which should be treated as a heuristic scaling limit, while the discrete $N(t)$ formulation requires the junction consistency already stated in Sec. IV. where $t_{\rm rec}$ is a reference transition time and H is the Hubble scale.

The multifractal scaling symmetry is specified as

$$x^\mu \mapsto \varphi\,x^\mu, \qquad N(t) \mapsto N(t) + 1, \qquad \psi \mapsto \psi/\varphi^{3/2}, \tag{40}$$

leading to the current

$$\mathcal{J}^\mu_n = \frac{\partial\mathcal{L}_n}{\partial(\partial_\mu\psi)}\,\delta\psi - \Big(\frac{\partial\mathcal{L}_n}{\partial(\partial_\mu\psi)}\,\partial_\nu\psi - \mathcal{L}_n\,g^\mu{}_\nu\Big)\delta x^\nu,$$
$$\nabla_\mu\mathcal{J}^\mu_n = 0. \tag{41}$$

and a corresponding conserved quantity (for a chosen time foliation)

$$E_n = \int d^3x\,\sqrt{-g_n}\;\mathcal{J}^0_n. \tag{42}$$

c. What is retained, and what is strengthened here. Equations (25)–(42) are the precise elements of the two preprints that are *needed* for the unified PRD manuscript: (i) the explicit symmetry map (27) and/or (40), (ii) the explicit current formulas (28) and (41), and (iii) the exact definition of the time-dependent embedding (33). The present manuscript goes beyond the sketches by (a) clarifying the on-shell nature of the current identities, (b) making junction conditions for floor-discretized $N(t)$ explicit (Theorem 2), and (c) separating *predictions* from *tests* via a likelihood protocol (Sec. V H).

Remark 2 (Covariance in the presence of a discrete embedding). The underlying gravitational sector remains diffeomorphism-invariant. In the *smooth* formulation one may treat $\nu(t) = \log_\varphi(ct/\ell_{\rm P})$ as an (effective) scalar embedding variable. In the *floor* formulation, $N(t)$ is piecewise-constant with a chosen foliation; adding the junction term and imposing the matching conditions ensure that the Euler–Lagrange system is distribution-free, so the Noether-II identity structure holds on each interval and is globally consistent across the transition hypersurfaces.

V. TOPOLOGY AND CMB OBSERVABLES

A. Invariant routes to the spherical sector

Diffeomorphism invariance in GR constrains the *local* variational structure via a Noether-II identity (Bianchi-type relations), but it does not by itself fix the *global* topology of Σ. To make the spherical sector mathematically unavoidable, one must add genuinely topological/geometric invariant conditions. We record two standard levels of "inevitability."

a. Hard inevitability (strict): $\Sigma \cong S^3$.

Theorem 3 (Hard spherical inevitability)**.** *Let Σ be a closed 3-manifold, i.e. compact and without boundary, $\partial\Sigma = \varnothing$. If Σ is simply connected,*

$$\pi_1(\Sigma) = 0, \tag{43}$$

then Σ is homeomorphic to the 3-sphere,

$$\Sigma \cong S^3. \tag{44}$$

This is the Poincaré/Geometrization criterion in dimension three [15–19].

b. Soft inevitability (minimal spherical): $\widetilde{\Sigma} \cong S^3$.

Corollary 2 (Soft spherical inevitability: spherical space forms)**.** *Let Σ be closed, $\partial\Sigma = \varnothing$. If Σ admits a Riemannian metric with strictly positive Ricci curvature,*

$$\mathrm{Ric} > 0, \tag{45}$$

then Σ is a spherical space form,

$$\Sigma \cong S^3/\Gamma, \qquad \widetilde{\Sigma} \cong S^3, \tag{46}$$

where Γ is a finite group acting freely by isometries [12–14].

In our observational pipeline we therefore treat $\Sigma \cong S^3/\Gamma$ as the relevant compact sector and compare candidate Γ via likelihood/evidence (Appendix A).

B. Current observational status of compact topology searches

Direct topology searches in the CMB target the existence of repeated patterns such as "matched circles" in the temperature/polarization sky maps [8]. Current analyses, including the Planck topology study, find no statistically significant detection and therefore provide lower bounds on the size (injectivity radius) of possible fundamental domains [9]. Recent dedicated limits on spherical lens spaces $L(p, q)$ from circle searches refine these bounds within the spherical sector S^3/Γ [10], and forecasts quantify the reach of future searches [11].

C. Curvature radius and mode discreteness

In a closed FRW slicing introduce curvature radius R_K; at present

$$R_{K,0} = \frac{c}{H_0\sqrt{|\Omega_k|}}, \qquad \Omega_k < 0, \tag{47}$$

On S^3 scalar harmonics satisfy

$$\Delta_{S^3} Y_{n\ell m} = -\frac{n(n+2)}{R_K^2}\, Y_{n\ell m}, \qquad n = 0, 1, 2, \ldots. \tag{48}$$

Define $k_n = (n+1)/R_K$.

D. A concrete toy numerical example: discrete S^3 mode sum

To make the discrete mode structure fully explicit (independently of any Boltzmann-code implementation), consider the scalar Laplacian spectrum (48). The associated "wavenumber" is $k_n = (n+1)/R_K$ and the scalar harmonic degeneracy on S^3 is $(n+1)^2$. Table I lists the first modes for an illustrative curvature radius $R_K = 50\,\mathrm{Gpc}$, and Fig. 1 visualizes the discretization (a uniform spacing $\Delta k = 1/R_K$). This is a minimal, checkable numerical instance of the replacement $k \to k_n$ used in (50).

E. Low-ℓ scale and C_ℓ template

With $\ell \sim k\chi_*$, the fundamental scale is

$$\ell_{\min} \sim \frac{\chi_*}{R_K}. \tag{49}$$

The angular spectrum becomes

$$C_\ell = 4\pi \sum_n \mathcal{P}_{\mathcal{R}}(k_n)\, |\Delta_\ell(k_n)|^2. \tag{50}$$

TABLE I. Toy S^3 scalar modes for $R_K = 50\,\mathrm{Gpc}$. Here $k_n = (n+1)/R_K$ and the scalar degeneracy is $(n+1)^2$.

n	k_n [Gpc^{-1}]	Δk [Gpc^{-1}]	degeneracy $(n+1)^2$
0	0.02	0.02	1
1	0.04	0.02	4
2	0.06	0.02	9
3	0.08	0.02	16
4	0.10	0.02	25
5	0.12	0.02	36
6	0.14	0.02	49
7	0.16	0.02	64
8	0.18	0.02	81
9	0.20	0.02	100
10	0.22	0.02	121

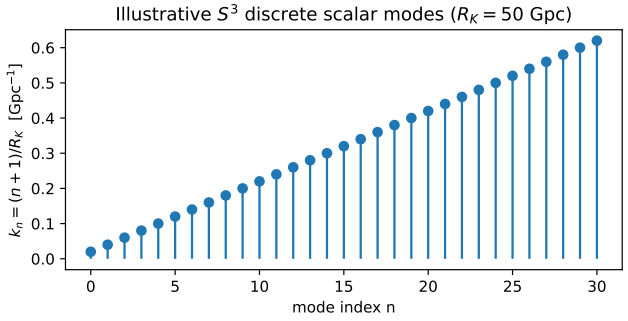


FIG. 1. Illustrative discretization of scalar modes on S^3 for $R_K = 50\,\mathrm{Gpc}$: $k_n = (n+1)/R_K$.

F. Matched circles for $\Sigma \simeq S^3/\Gamma$

If $\Sigma \simeq S^3/\Gamma$ with nontrivial Γ, matched-circle correlations [8, 9] may appear; non-detection yields constraints on the fundamental domain size.

G. Minimal Planck likelihood pipeline

1. Choose (R_K, Ω_k) and $\mathcal{P}_\mathcal{R}(k)$.
2. Compute closed-geometry transfer functions and evaluate (50).
3. Fit low-ℓ likelihood and constrain R_K (and embedding selection parameters).
4. Apply matched-circle constraints if $\Gamma \neq 1$.

H. Likelihood protocol for observational verification

The two companion preprints contained *qualitative* "alignment" statements with CMB and GW data. For PRD-grade verification we adopt an explicit likelihood protocol. The minimal requirement is that the model be implemented as a prediction map $\theta \mapsto \{C_\ell, \Omega_{\rm GW}(f), \dots\}$ and tested against public likelihoods.

a. CMB. Let $\mathbf{D}$ denote the binned Planck [20, 21] data vector (e.g. TT,TE,EE) and $\mathbf{M}(\theta)$ the model prediction obtained from a Boltzmann code (CLASS/CAMB) [22, 23] in a closed geometry, with the replacement $k \to k_n = (n+1)/R_K$ in the mode sums (50) and a parameterization of the primordial spectrum $\mathcal{P}_\mathcal{R}(k)$ that allows for the φ-indexed features. The (Gaussian-approximate) log-likelihood reads

$$-2\ln\mathcal{L}_{\rm CMB}(\theta) = \big(\mathbf{D}-\mathbf{M}(\theta)\big)^\top \mathbf{C}^{-1}\big(\mathbf{D}-\mathbf{M}(\theta)\big), \quad (51)$$

with covariance $\mathbf{C}$ provided by the likelihood package. Model selection uses $\Delta\chi^2 = -2\Delta\ln\mathcal{L}$ against the ΛCDM baseline at matched nuisance treatment.

b. Implementation notes (CLASS/CAMB + Cobaya/MontePython). A minimal reproducible pipeline is: (i) implement the discrete-k sums and any φ-indexed feature parameters in a closed-geometry Boltzmann solver (CLASS/CAMB) [22, 23]; (ii) couple to an MCMC / nested-sampling front-end (Cobaya or MontePython) [24, 25] using the official Planck likelihoods; (iii) publish the full chain set and posterior plots with a standard analyzer (e.g. GetDist) [26].

c. Stochastic gravitational-wave background. Given a prediction $\Omega_{\rm GW}(f;\theta)$ (Sec. VI), compare to LVK [27] and PTA constraints via their published likelihoods. A simple summary statistic is

$$\chi^2_{\rm GW}(\theta) = \sum_i \frac{\big(\Omega_{\rm GW}(f_i;\theta) - \Omega_i^{\rm obs}\big)^2}{\sigma_i^2}, \quad (52)$$

for an appropriate set of frequency bins $(f_i, \Omega_i^{\rm obs}, \sigma_i)$, while the definitive test uses the collaboration likelihoods.

d. Non-negotiable deliverables. A complete submission must therefore include: (i) a reproducible code path generating C_ℓ and $\Omega_{\rm GW}(f)$ from the manuscript equations, (ii) a parameter table with priors, and (iii) a $\Delta\chi^2$ (or Bayes factor) report against standard baselines.

VI. STOCHASTIC GRAVITATIONAL-WAVE BACKGROUND

A. Discrete tensor spectrum and frequencies

Tensor harmonics on S^3 form a discrete set:

$$\Delta_{S^3} T_{ij}^{(n)} = -\frac{\lambda_n^{(T)}}{R_K^2}\, T_{ij}^{(n)}, \qquad n \geq 3, \quad (53)$$

with $k_n^{(T)} \sim \sqrt{\lambda_n^{(T)}}/R_K$ and $f_n = \frac{c}{2\pi}\frac{k_n^{(T)}}{a_0}$.

B. Definition and qualitative signature of $\Omega_{\rm GW}(f)$

$$\Omega_{\rm GW}(f) \equiv \frac{1}{\rho_c}\frac{{\rm d}\rho_{\rm GW}}{{\rm d}\ln f}, \qquad \rho_c = \frac{3H_0^2c^2}{8\pi G}. \tag{54}$$

Discreteness yields (schematically)

$$\Omega_{\rm GW}(f) \sim \sum_{n\geq 3} \mathcal{P}_T(k_n^{(T)})\,\mathcal{T}_T^2(k_n^{(T)})\,\delta(f-f_n), \tag{55}$$

implying an IR cutoff and possible step/comb features set by $1/(a_0R_K)$.

Equation (55) is schematic: in any realistic analysis the δ-comb is broadened by finite observation time, redshift smearing, and the bandwidth of estimators used in PTA/LVK searches. Moreover, for curvature radii compatible with current near-flat constraints, the mode spacing in the PTA/LVK bands is effectively continuous: one has $\Delta f/f \sim 1/n$ with $n \sim 2\pi f\, a_0R_K/c \gg 1$. Current PTA detections are well-fit by a smooth power-law spectrum consistent with astrophysical SMBHB populations [28, 29]; therefore topology-induced discreteness must either lie in a marginally allowed regime (very large R_K) or appear only as weak modulations on top of a smooth background.

C. Minimal SGWB verification protocol

1. Fix R_K and specify $\mathcal{P}_T(k)$.
2. Compute $(k_n^{(T)}, f_n)$.
3. Build broadened $\Omega_{\rm GW}(f)$ prediction.
4. Compare to PTA and ground-based constraints [27–29]; forecast detectability of topology-induced features.

VII. CONCLUSION

Diffeomorphism invariance yields Noether-II identities rather than global Noether-I charges. Associated covariant charges are boundary charges and vanish on compact boundaryless Σ (notably S^3). Embedding-level modifications are variationally well-defined either as smooth $\nu(t)$ dependence or as discrete floor $N(t)$ with explicit junction consistency. Compact topology implies discrete mode spectra, giving falsifiable signatures in low-ℓ CMB and infrared SGWB, testable through likelihood analyses.

Present data already impose strong constraints on both spatial curvature and compact topology. In particular, Planck cosmological parameter analyses tightly constrain curvature around $\Omega_k \approx 0$ [21], while dedicated "matched circles" and related searches constrain the size of many compact fundamental domains to be comparable to or larger than the surface of last scattering [9–11]. Accordingly, the discrete-mode signatures emphasized here are expected to appear only in the marginally allowed tail of parameter space (very large R_K and/or nearly-horizon fundamental domains), with upcoming CMB and PTA datasets providing decisive tests.

a. Invariant–constraint map. To make the falsifiable content explicit, we summarize the chain *invariant* ⇒ *observable* ⇒ *present constraint* ⇒ *likelihood module*:

1. **Diffeomorphism invariance ⇒ Noether-II identities and boundary charges.** In generally covariant theories the on-shell diffeomorphism charge is a boundary integral [1, 5]. Observable: vanishing global diffeomorphism charges on compact boundaryless slices ($\partial\Sigma = \varnothing$). Constraint/module: the analysis is implemented at the level of covariant identities and requires no asymptotic boundary data.

2. **Spherical space form sector $\widetilde{\Sigma} \simeq S^3$ ⇒ discrete Laplacian spectra.** For $\Sigma \simeq S^3/\Gamma$ the scalar/tensor mode spectra are discrete [12, 13]. Observable: low-ℓ discreteness/cutoff in C_ℓ. Constraint/module: tested against Planck likelihood and curvature constraints [20, 21] and topology searches [9–11].

3. **Invariant refinement bookkeeping ⇒ unique scale ladder ⇒ φ.** Requiring unambiguous composition of refinement steps fixes the Zeckendorf–Lekkerkerker encoding and hence the Fibonacci ladder, with φ as the unique positive fixed point of the induced ratio map [33, 34]. Observable: step/log-periodic features tied to $N(t)$ transitions. Constraint/module: junction conditions enforce absence of spurious distributional sources (Sec. IV).

4. **Compactness + tensor discreteness ⇒ IR structure in $\Omega_{\rm GW}(f)$.** Observable: broadened IR features (rather than idealized delta-combs) in the PTA/LVK bands. Constraint/module: compared to PTA and ground-based bounds [27–29].

VIII. VERIFICATION CHECKLIST

1. Implement closed-geometry mode functions; compute C_ℓ from (50).
2. Fit Planck low-ℓ likelihood; constrain R_K and topology parameters.
3. Implement tensor discreteness; compare $\Omega_{\rm GW}(f)$ to PTA and ground-based bounds.
4. Report likelihood metrics (χ^2, Bayes factors), not qualitative claims.

Appendix A: From invariant assumptions to S^3: hard vs. soft inevitability

1. Invariant, sector, and test

In this paper, "invariant" means coordinate-free and gauge-independent. Diffeomorphism invariance in GR yields Noether-II identities that constrain the local form of the field equations and the boundary-term structure, but it does not uniquely fix global topology. A global assumption such as compactness and boundarylessness ($\partial\Sigma = \varnothing$) specifies a topological sector of configuration space, while additional topological/geometric conditions can make the spherical sector mathematically unavoidable.

Hard inevitability corresponds to the topological invariant $\pi_1(\Sigma) = 0$, forcing $\Sigma \cong S^3$ (Theorem 3). Soft inevitability corresponds to the geometric invariant $\mathrm{Ric} > 0$, forcing spherical space forms $\Sigma \cong S^3/\Gamma$ with $\widetilde{\Sigma} \cong S^3$ (Corollary 2). In practice, present data typically constrain the size and group label Γ rather than proving trivial Γ; hence we implement explicit likelihood comparison within the spherical sector.

2. Likelihood selection within the spherical sector

Within $\Sigma \cong S^3/\Gamma$, topology enters observables through the Laplacian eigenmodes and the induced correlation structure of perturbations (and their φ-refined discretization). This enables model comparison via Bayesian evidence and Bayes factors:

$$\mathcal{Z}_\Gamma = \int d\theta\, \mathcal{L}(D \mid \theta, \Gamma)\, \pi(\theta \mid \Gamma), \qquad B_{\Gamma_1\Gamma_2} = \frac{\mathcal{Z}_{\Gamma_1}}{\mathcal{Z}_{\Gamma_2}}. \quad \text{(A1)}$$

Here Γ labels the (finite) spherical space-form group and θ collects cosmological and nuisance parameters. This is the operational implementation of the invariant $\to$ constraint $\to$ observable $\to$ likelihood map used throughout the manuscript.

Appendix B: Invariant sieve to S^3: properties, theorems, and experimental handles

The goal of this appendix is to make explicit the "invariant sieve" logic used in the main text. We start from a broad candidate class of spatial topologies Σ (closed, smooth 3-manifolds), and we collect coordinate-free and gauge-independent inputs ("invariants") that translate into topological or geometric constraints. The admissible class is then the intersection

$$\mathfrak{T}_{\mathrm{adm}} = \bigcap_i \mathcal{C}_i, \qquad \mathcal{C}_i \subset \mathfrak{T}_0 \ \text{(candidate class)}, \quad \text{(B1)}$$

where each $\mathcal{C}_i$ is induced by an invariant statement and/or a data-driven constraint. The sieve is strongest when (i) the mathematical implication "invariant $\Rightarrow$ constraint" is theorem-level, and (ii) the associated observable is actually testable by a concrete likelihood analysis.

What can and cannot be "uniquely forced" by experiment. Current cosmological topology searches primarily provide bounds on the size of the fundamental domain and exclude specific nontrivial identifications up to some scale, rather than proving uniqueness of $\pi_1(\Sigma)$ or of the group label Γ [8, 9]. Accordingly, the sieve has two layers: (a) a theorem-driven layer (hard/soft inevitability) that can uniquely single out S^3 under explicit topological/geometric invariants, and (b) a likelihood layer that constrains (or rules out) candidate quotients S^3/Γ.

Sieve conclusion. The combined effect of (i) Noether-consistency in the compact sector (boundary-charge structure), (ii) theorem-level hard/soft routes to the spherical sector, and (iii) explicit likelihood comparison over Γ via topology-dependent mode spectra, is to reduce the admissible topology class to the spherical sector S^3/Γ and to treat S^3 as the minimal representative. A mathematically unique identification $\Sigma \cong S^3$ is obtained once the hard invariant $\pi_1(\Sigma) = 0$ is established or adopted as a required constraint; observationally, this corresponds to progressively excluding nontrivial identifications through topology searches and likelihood-based model comparison.

Property of S^3 (invariant)	**Theorem-level implication**	**Experimental / observational handle**
Closed and boundaryless ($\partial\Sigma = \varnothing$)	In GR, diffeo-charges are boundary charges; for $\partial\Sigma = \varnothing$ the corresponding global boundary integrals vanish (Sec. IV F). This fixes the correct meaning of "conservation" in the compact sector.	Not directly measurable as a binary fact; motivates focusing on topology-dependent correlation structure rather than global energy.
Simply connected ($\pi_1(S^3) = 0$)	*Hard inevitability:* for a closed 3-manifold, $\pi_1(\Sigma) = 0$ implies $\Sigma \cong S^3$ (Poincaré/Perelman; used as Theorem 3).	CMB topology searches (matched circles and related correlation tests) constrain nontrivial identifications and hence possible nontrivial $\pi_1(\Sigma)$ at scales comparable to the LSS radius [8, 9].
Spherical space-form sector (constant $K > 0$)	If Σ admits a metric of constant positive sectional curvature then $\Sigma \cong S^3/\Gamma$ with $\widetilde{\Sigma} \cong S^3$ (spherical space-form classification; used in Corollary 2 as sector target).	CMB topology constraints restrict specific Γ and the fundamental-domain size; likelihood analysis compares Γ-labels via mode spectra and correlation structure (Appendix A 2).
Positive Ricci curvature route (Ric > 0)	*Soft inevitability:* Ric > 0 on a closed 3-manifold forces a spherical space form S^3/Γ (Appendix A).	Not directly measured as a geometric invariant of spatial slices; implemented as a physically motivated sector prior.
Discrete Laplacian spectrum on S^3	Compactness implies a discrete spectrum; on S^3 the eigenmodes are hyperspherical harmonics, enabling replacement of $\int dk$ by a mode sum (Sec. V B, Eq. (50)).	Low-ℓ CMB power and correlation anomalies are sensitive to IR mode content; topology searches and Planck likelihood constrain topological parameters and curvature scale [9].
Spin structure / global fermions	Any oriented 3-manifold is spin (vanishing w_2); S^3 is spin and admits globally defined spinors, consistent with including fermionic quantum fields.	Laboratory quantum experiments show that topological information can be encoded in gauge-invariant phases and discrete indices (e.g. Aharonov–Bohm and quantum Hall invariants)[30–32]. This supports "topology can be observable" but does not by itself fix cosmic Σ.

TABLE II. Invariant sieve: representative topological/geometric properties of S^3, theorems that map them to restrictions on admissible Σ, and the corresponding experimental/observational handles. Theorem-level inputs (hard/soft) can single out the spherical sector, while data-driven searches constrain the quotient label Γ and the fundamental-domain size within that sector.

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